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SEMICONDUCTOR DEVICES AND MANUFACTURING METHODS OF THE SAME

Abstract

A method for manufacturing a semiconductor device includes forming a first substrate structure including a first substrate, first memory cells disposed on the first substrate, first bit lines disposed on the first memory cells and connected to the first memory cells, and first bonding pads disposed on the first bit lines to be connected to the first bit lines, respectively, forming a second substrate structure including a second substrate, second memory cells disposed on the second substrate, second bit lines disposed on the second memory cells and connected to the second memory cells, and second bonding pads disposed on the second bit lines to be connected to the second bit lines, respectively, and bonding the second substrate structure to the first substrate structure by bonding the first bonding pads and the second bonding pads. The first bit lines are electrically connected to the second bit lines, respectively, through the first bonding pads and the second bonding pads, and the first bonding pads and second bonding pads are vertically between the first bit lines and the second bit lines.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application of U.S. patent application Ser. No. 17/510,594, filed Oct. 26, 2021, which is a continuation application of U.S. patent application Ser. No. 16/397,556, filed Apr. 29, 2019, which claims priority to Korean Patent Application No. 10-2018-0116804 filed on Oct. 1, 2018 in the Korean Intellectual Property Office, the disclosure of each of which is incorporated herein by reference in its entirety.

BACKGROUND
1. Field
[0002] The present inventive concept relates to a semiconductor device and a method for manufacturing the same.
2. Description of Related Art
[0003] Semiconductor devices are increasingly required to process high-capacity data while being gradually reduced in volume. Correspondingly, there is a need to increase a degree of integration of semiconductor elements forming such semiconductor devices. Resultantly, as one method of increasing a degree of integration of semiconductor elements, a semiconductor device having a vertical transistor structure, in place of a conventional planar transistor structure, has been proposed.
SUMMARY
[0004] An aspect of the present inventive concept is to provide a semiconductor device having improved degrees of integration and reliability and a method for manufacturing the same.
[0005] According to an aspect of the present inventive concept, a semiconductor device includes a first substrate structure and a second substrate structure. The first substrate structure includes a first substrate, first gate electrodes stacked and spaced apart from each other in a direction perpendicular to a first surface of the first substrate, first channels extending perpendicular to the first substrate while passing through the first gate electrodes, first bit lines connected to the first channels, and first bonding pads disposed on the first bit lines to be electrically connected to the first bit lines,

wherein, in the direction perpendicular to the first surface of the first substrate, the first bit lines are disposed between the first channels and the first bonding pads, and the first channels extend between the first substrate and the first bit lines. The second substrate structure is connected to the first substrate structure on the first substrate structure, and includes a second substrate, second gate electrodes stacked and spaced apart from each other in a direction perpendicular to a first surface of the second substrate that faces the first surface of the first substrate, second channels extending perpendicular to the second substrate while passing through the second gate electrodes, second bit lines connected to the second channels, and second bonding pads disposed on the second bit lines to be electrically connected to the second bit lines, wherein, in the direction perpendicular to the first surface of the second substrate, the second bit lines are disposed between the second channels and the second bonding pads, and the second channels extend between the second substrate and the second bit lines. The first substrate structure and the second substrate structure are bonded together by the first bonding pads and the second bonding pads and connected to each other, and the first bit lines are electrically connected to the second bit lines, respectively, through the first bonding pads and the second bonding pads.

[0006] According to an aspect of the present inventive concept, which may be the same embodiment as the aforementioned aspect or a different embodiment, the first substrate structure includes a first substrate, first gate electrodes stacked and spaced apart from each other in a direction perpendicular to a first surface of the first substrate and extended by different lengths in one direction to provide first contact regions, first channels extending perpendicular to the first substrate while passing through the first gate electrodes, first cell contact plugs connected to the first gate electrodes in the first contact regions and extending perpendicular to the first surface of the first substrate, first bit lines connected to the first channels, and first bonding pads disposed at a first surface of the first substrate structure, wherein the first bit lines are disposed between the first channels and the first bonding pads in a direction perpendicular to the first surface of the first substrate. The second substrate structure may be connected to the first substrate structure on the first substrate structure, and may include a second substrate, second gate electrodes stacked and spaced apart from each other in a direction perpendicular to a first surface of the second substrate and extended by different lengths in one direction to provide second contact regions, second channels extending perpendicular to the second substrate while passing through the second gate electrodes, second cell contact plugs connected to the second gate electrodes in the second contact regions and extending perpendicular to the first surface of the second substrate, second bit lines connected to the second channels, and second bonding pads bonded to the first bonding pads and disposed at a first surface of the second substrate structure. The first surface of the first substrate faces the first surface of the second substrate, and the first surface of the first substrate structure faces the first surface of the second substrate structure, and the first bit lines are electrically connected to the second bit lines through respective bonding pads of the first bonding pads and the second bonding pads, and some of the first cell contact plugs are respectively electrically connected to the second cell contact plugs through respective bonding pads of the first bonding pads and the second bonding pads.

[0007] According to an aspect of the present inventive concept, which may be the same embodiment as the aforementioned aspects, or a different embodiment, the first substrate structure includes a base substrate, circuit elements disposed on the base substrate, a first substrate disposed on the circuit elements, first memory cells disposed on the first substrate and electrically connected to the circuit elements, first bit lines disposed on the first memory cells and connected to the first memory cells, and first bonding pads disposed on the first bit lines to be connected to the first bit lines, respectively. The second substrate structure is connected to the first substrate structure on the first substrate structure, and includes a second substrate, second memory cells disposed on the second substrate, second bit lines disposed on the second memory cells and connected to the second memory cells, and second bonding pads disposed on the second bit lines to be connected to the second bit lines, respectively. The first substrate structure and the second substrate structure are connected to each other by bonding the first bonding pads to the second bonding pads, and the first bonding pads and second bonding pads are vertically between the first bit lines and the second bit lines, without the first substrate or second substrate disposed vertically between the first bit lines and the second bit lines.

[0008] According to an aspect of the present inventive concept, a method for manufacturing a semiconductor device includes: forming a first substrate structure by forming first gate electrodes stacked and spaced apart from each other in a direction perpendicular to a first surface of a first substrate, first channels extending perpendicular to the first substrate while passing through the first gate electrodes, first bit lines connected to the first channels, and first bonding pads disposed on the first bit lines to be electrically connected to the first bit lines, respectively, on the first substrate; forming a second substrate structure by forming second gate electrodes stacked and spaced apart from each other in a direction perpendicular to a first surface of a second substrate, second channels extending perpendicular to the second substrate while passing through the second gate electrodes, second bit lines connected to the second channels, and second bonding pads disposed on the second bit lines to be electrically connected to the second bit lines, respectively, on the second substrate; forming a third substrate structure by forming circuit elements, through contact plugs passing through the third substrate to a predetermined depth, and third bonding pads disposed on the circuit elements, on a first surface of the third substrate; bonding the third substrate structure to the first substrate structure by bonding the first bonding pads to the third bonding pads; exposing first ends of the through contact plugs by removing a portion of the third substrate from a second surface of the third substrate, opposite the first surface; forming fourth bonding pads on the through contact plugs, exposed through the second surface of the third substrate; and bonding the second substrate structure to the third substrate structure by bonding the second bonding pads to the fourth bonding pads.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] The above and other aspects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:

[0010] FIG. **1** is a schematic block diagram of a semiconductor device according to example embodiments;

[0011] FIGS. **2**A and **2**B are equivalent circuit diagrams of a cell array of a semiconductor device according to example embodiments;

[0012] FIGS. **3**A and **3**B are schematic layout diagrams illustrating an arrangement of a semiconductor device according to example embodiments;

[0013] FIG. **4** is a schematic plan view of a semiconductor device according to example embodiments;

[0014] FIG. **5** is a schematic cross-sectional view of a semiconductor device according to example embodiments;

[0015] FIG. **6** is a layout diagram illustrating a portion of a semiconductor device according to example embodiments;

[0016] FIGS. **7**A and **7**B are partially enlarged views of a semiconductor device according to example embodiments;

[0017] FIGS. **8**A and **8**B are partially enlarged views of a semiconductor device according to example embodiments;

[0018] FIGS. **9** to **12** are schematic cross-sectional views of a semiconductor device according to

example embodiments;
[0019] FIG. **13** is a schematic cross-sectional view of a semiconductor device according to example embodiments;
[0020] FIGS. **14**A to **14**H are schematic cross-sectional views illustrating a method for manufacturing a semiconductor device according to example embodiments;
[0021] FIGS. **15**A to **15**G are schematic cross-sectional views illustrating a method for manufacturing a semiconductor device according to example embodiments;
[0022] FIG. **16** is a block diagram illustrating an electronic device including a semiconductor device according to example embodiments.
DETAILED DESCRIPTION
[0023] Hereinafter, the example embodiments of the present disclosure will be described in detail with reference to the attached drawings. In the following description, terms such as ‘upper,’ ‘upper portion,’ ‘upper surface,’ ‘lower,’ ‘lower portion,’ ‘lower surface,’ and ‘side surface,’ and the like may be understood to refer to the reference to the drawings, unless otherwise indicated by reference numerals and referred to separately.
[0024] It will be understood that when an element is referred to as being “connected” or “coupled” to or “on” another element, it can be directly connected or coupled to or on the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, or as “contacting” or “in contact with” another element, there are no intervening elements present at the point or points of contact or connection. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.).
[0025] FIG. **1** is a schematic block diagram of a semiconductor device according to example embodiments.
[0026] Referring to FIG. **1**, a semiconductor device **10** may include a memory cell array **20** and a peripheral circuit **30**. The peripheral circuit **30** may include a row decoder **32**, a page buffer **34**, an input/output (I/O) buffer **35**, a control logic **36**, and a voltage generator **37**.
[0027] The memory cell array **20** may include a plurality of memory blocks, and each of the memory blocks may include a plurality of memory cells. The plurality of memory cells may be connected to the row decoder **32** through a string select line SSL, word lines WL, and a ground select line GSL, and may be connected to the page buffer **34** through bit lines BL. In example embodiments, a plurality of memory cells arranged in an identical row may be connected to an identical word line WL, and a plurality of memory cells arranged in an identical column may be connected to an identical bit line BL.
[0028] The row decoder **32** may decode an address ADDR, having been input, and may thus generate and transmit driving signals of the word line WL. The row decoder **32** may provide a word line voltage, generated from the voltage generator **37**, to a selected word line WL and unselected word lines WL, in response to control of the control logic **36**.
[0029] The page buffer **34** is connected to the memory cell array **20** through the bit lines BL, and thus reads information stored in the memory cells. The page buffer **34** may temporarily store data to be stored in the memory cells, or may sense data, stored in the memory cell, according to a mode of operation. The page buffer **34** may include a column decoder and a sense amplifier. The column decoder may selectively activate bit lines BL of the memory cell array **20**, while the sense amplifier may sense a voltage of a bit line BL, selected by the column decoder, and may thus read data, stored in a memory cell, having been selected.
[0030] The I/O buffer **35** may receive data DATA and transfer the data to the page buffer **34** during a programming operation, and may output the data DATA, transferred by the page buffer **34**, externally, during a reading operation. The I/O buffer **35** may transmit an address or command, having been input, to the control logic **36**.

[0031] The control logic **36** may control operations of the row decoder **32** and the page buffer **34**. The control logic **36** may receive a control signal and an external voltage, transmitted from an external source, and may be operated according to a control signal, having been received. The control logic **36** may control reading, writing, and/or erasing operations in response to the control signals.

[0032] The voltage generator **37** may generate voltages, for example, a programming voltage, a reading voltage, an erasing voltage, and the like, required for an internal operation using an external voltage. The voltage, generated by the voltage generator **37**, may be transferred to the memory cell array **20** through the row decoder **32**.

[0033] FIGS. **2**A and **2**B are equivalent circuit diagrams of a cell array of a semiconductor device according to example embodiments.

[0034] Referring to FIG. **2**A, a memory cell array **20**A may include a plurality of first memory cell strings CS**1**, each of which includes first memory cells MC**1** connected to each other in series, and a first ground select transistor GST**1** and first string select transistors SST**1_1** and SST**1_2** connected to both ends of the first memory cells MC**1** in series. The plurality of first memory cell strings CS**1** may be connected to respective first bit lines BL**1_0** to BL**1_2** in parallel. The plurality of first memory cell strings CS**1** may be connected to a first common source line CSL**1** in common. In other words, the plurality of first memory cell strings CS**1** may be disposed between the plurality of first bit lines BL**1_0** to BL**1_2** and a single first common source line CSL**1**. In an example embodiment, a plurality of first common source lines CSL**1** may be arranged two-dimensionally.

[0035] Moreover, the memory cell array **20**A may include a plurality of second memory cell strings CS**2**, each of which includes second memory cells MC**2** disposed on an upper portion of the first bit lines BL**1_0** to BL**1_2**, and connected to each other in series, and a second ground select transistor GST**2** and second string select transistors SST**2_1** and SST**2_2** connected to both ends of the second memory cells MC**2** in series. The plurality of second memory cell strings CS**2** may be connected to respective second bit lines BL**2_0** to BL**2_2** in parallel. The plurality of second memory cell strings CS**2** may be connected to a second common source line CSL**2** in common. As shown, the plurality of second memory cell strings CS**2** may be disposed between the plurality of second bit lines BL**2_0** to BL**2_2** and a single second common source line CSL**2**.

[0036] The first bit lines BL**1_0** to BL**1_2** and the second bit lines BL**2_0** to BL**2_2**, vertically disposed in the memory cell array **20**A, may be electrically connected to each other. The first memory cell strings CS**1** and the second memory cell strings CS**2** may have substantially the same circuit structure, based on the first bit lines BL**1_0** to BL**1_2** and the second bit lines BL**2_0** to BL**2_2**. In the first memory cell strings CS**1** and the second memory cell strings CS**2**, the first string select lines SSL**1_1** and SSL**1_2** and the second string select lines SSL**2_1** and SSL**2_2** may be electrically connected to each other and may be in an equipotential state, and the first ground select line GSL**1** and the second ground select line GSL**2** may also be electrically connected to each other and may be in an equipotential state. However, different signals may be applied to the first word lines WL**1_0** to WL**1**_*n* and the second word lines WL**2_0** to WL**2**_*n*. Thus, different types of data may be written on the first memory cells MC**1** and the second memory cells MC**2**, respectively. Hereinafter, a description common to the first memory cell strings CS**1** and the second memory cell strings CS**2** will be described together without distinguishing of the first memory cell strings CS**1** and the second memory cell strings CS**2**.

[0037] The memory cells MC**1** and MC**2**, connected to each other in series, may be controlled by word lines WL**1_0** to WL**1**_*n* and WL**2_0** to WL**2**_*n* for selecting the memory cells MC**1** and MC**2**. Each of the memory cells MC**1** and MC**2** may include a data storage element. Gate electrodes of the memory cells MC**1** and MC**2**, arranged at substantially the same distance from the common source lines CSL**1** and CSL**2**, may be commonly connected to one of the word lines WL**1_0** to WL**1**_*n* and WL**2_0** to WL**2**_*n* and may be in an equipotential state. Alternatively, even when the gate electrodes of the memory cells MC**1** and MC**2** are arranged at substantially the same distance from the common source lines CSL**1** and CSL**2**, gate electrodes, disposed in different rows or columns, may be controlled independently.

[0038] The ground select transistors GST**1** and GST**2** may be controlled by the ground select lines GSL**1** and GSL**2**, and may be connected to the common source lines CSL**1** and CSL**2**. The string select transistors SST**1_1**, SST**1_2**, SST**2_1**, and SST**2_2** may be controlled by the string select lines SSL**1_1**, SSL**1_2**, SSL**2_1**, and SSL**2_2**, and may be connected to the bit lines BL**1_0** to BL**1_2** and BL**2_0** to BL**2_2**. FIG. **2**A illustrates a structure in which a single ground select transistor GST**1** and GST**2** and two string select transistors SST**1_1**, SST**1_2**, SST**2_1**, and SST**2_2** are connected to the plurality of memory cells MC**1** and MC**2** connected to each other in series, respectively. In a different manner, a single string select transistor, or a plurality of ground select transistors may also be connected to the memory cells. One or more dummy lines DWL**1** and DWL**2** or buffer lines may be further disposed between an uppermost word line WL**1**_*n* and WL**2**_*n*, among the word lines WL**1_0** to WL**1**_*n* and WL**2_0** to WL**2**_*n*, and the string select lines SSL**1_1**, SSL**1_2**, SSL**2_1**, and SSL**2_2**. In an example embodiment, one or more dummy lines DWL**1** and DWL**2** may also be disposed between a lowermost word line WL**1_0** and WL**2_0** and the ground select line GSL**1** and GSL**2**. In the present specification, elements referred to by the term ‘dummy’ may have the same or a similar structure and shape to that of other components, and may only be used to refer to a component present as a pattern, without a practical function within a device (e.g., it may be connected to memory cells whose stored information is ignored by a host or controller).

[0039] When a signal is applied to the string select transistors SST**1_1**, SST**1_2**, SST**2_1**, and SST**2_2** through the string select lines SSL**1_1**, SSL**1_2**, SSL**2_1**, and SSL**2_2**, a signal, applied through the bit lines BL**1_0** to BL**1_2** and BL**2_0** to BL**2_2**, may be transmitted to the memory cells MC**1** and MC**2**, connected to each other in series, and a data reading operation and a data writing operation may be performed. Moreover, a predetermined erasing voltage is applied through a substrate, so an erasing operation for erasing data, written on the memory cells MC**1** and MC**2**, may be performed. In an example embodiment, the memory cell array **20**A may include at least one dummy memory cell string, electrically isolated from the bit lines BL**1_0** to BL**1_2** and BL**2_0** to BL**2_2**.

[0040] As can be seen in FIG. **2**A, in one embodiment, the memory cell array **10**A may have a bilaterally symmetric structure around a portion of the array where bit lines BL**1_0** to BL**1_2** and BL**2_0** to BL**2_2** connect to each other.

[0041] Referring to FIG. **2**B, the first bit lines BL**1_0** to BL**1_2** and the second bit lines BL**2_0** to BL**2_2**, vertically disposed in a memory cell array **20**B, may be electrically connected to each other. In a manner similar to that illustrated in FIG. **2**A, the first memory cell strings CS**1** and the second memory cell strings CS**2** may have substantially the same circuit structure, based on the first bit lines BL**1_0** to BL**1_2** and the second bit lines BL**2_0** to BL**2_2**. However, in a manner different from that illustrated in FIG. **2**A, in the first memory cell strings CS**1** and the second memory cell strings CS**2**, the first word lines WL**1_0** to WL**1**_*n* and the second word lines WL**2_0** to WL**2**_*n* may be electrically connected to each other and may be in equipotential state. Moreover, the first ground select line GSL**1** and the second ground select line GSL**2** are also electrically connected to each other and may be in equipotential state. On the other hand, the first string select line SSL**1_1** and SSL**1_2** and the second string select line SSL**2_1** and SSL**2_2** may be separately controlled by applying different signals. Thus, different types of data may be written on the first memory cells MC**1** and the second memory cells MC**2**, respectively. On the other hand, in example embodiments, the first string select line SSL**1_1** and SSL**1_2** and the second string select line SSL**2_1** and SSL**2_2** may be electrically connected to each other. In this case, the first memory cell strings CS**1** and the second memory cell strings CS**2** may be operated in the same manner, and data may be written and erased in the first memory cells MC**1** and the second memory cells MC**2** in the same manner.

[0042] FIGS. **3**A and **3**B are schematic layout diagrams illustrating an arrangement of a semiconductor device according to example embodiments.
[0043] Referring to FIG. **3**A, a semiconductor device **10**A may include a first substrate structure S**1** and a second substrate structure S**2**, stacked in a vertical direction. The first substrate structure S**1** may include a first region R**1** and a second region R**2**, the first region R**1** may form the peripheral circuit **30** of FIG. **1**, and the second region R**2** may form the memory cell array **20**. The first region R**1** may include a row decoder DEC, a page buffer PB, and other peripheral circuits PERI. The second region R**2** may include first memory cell arrays MCA**1** and a through wiring region TB. The second substrate structure S**2** may form the memory cell array **20**, and may include second memory cell arrays MCA**2**. In some embodiments, the first region R**1** is a first vertical region spanning a first vertical height range, and the second region R**2** is a second vertical region spanning a second vertical height range. The first region R**1** may be referred to as a first level, or bottom level, the second region R**2** may be referred to as a second level, or middle level, and the second substrate structure S**2** may be described as being at a third level, or top level.
[0044] In the first region R**1**, the row decoder DEC corresponds to the row decoder **32** described above with reference to FIG. **1**, and the page buffer PB may correspond to a region corresponding to the page buffer **34**. Moreover, other peripheral circuits PERI may be a region including the control logic **36** and the voltage generator **37** of FIG. **1**, and may include, for example, a latch circuit, a cache circuit, or a sense amplifier. In addition, other peripheral circuits PERI may include the I/O buffer **35** of FIG. **1**, and may include an electrostatic discharge (ESD) element or a data input/output circuit. In example embodiments, the I/O buffer **35** may be disposed to form a separate region around other peripheral circuits PERI.
[0045] In the first region R**1**, at least a portion of various circuit regions DEC, PB, and PERI described above may be disposed in a lower portion of the first memory cell arrays MCA**1** of the second region R**2**. For example, the page buffer PB and other peripheral circuits PERI may be disposed to overlap the first memory cell arrays MCA**1** in a lower portion of the first memory cell arrays MCA**1**. However, in example embodiments, circuits included in the first region R**1** and arrangement may be variously changed, so circuits disposed to overlap the first memory cell arrays MCA**1** may also be variously changed.
[0046] In the second region R**2**, the first memory cell arrays MCA**1** may be spaced apart from each other and disposed in parallel. However, in example embodiments, the number of the first memory cell arrays MCA**1** disposed in the first region R**2** and arrangement may be variously changed. For example, the first memory cell arrays MCA**1** according to the example embodiment may be arranged continuously and repeatedly.
[0047] The through wiring regions TB may be a region including a wiring structure passing through the second region R**2** and connected to the first region R**1**. The through wiring regions TB may be disposed on at least one side of the first memory cell arrays MCA**1**, and may include, for example, a wiring structure such as a contact plug electrically connected to the row decoder DEC of the first region R**1**, or the like. However, a through wiring structure may be disposed in the first memory cell arrays MCA**1**. For example, regions including a wiring structure electrically connected to the page buffer PB of the first region R**1** may be disposed therein.
[0048] In the second substrate structure S**2**, the second memory cell arrays MCA**2** may be spaced apart from each other and disposed in parallel. The second memory cell arrays MCA**2** may be disposed in a position corresponding to the first memory cell arrays MCA**1** of the first substrate structure S**1**, but are not limited thereto. In example embodiments, the number of the second memory cell arrays MCA**2** disposed in the second substrate structure S**2** and arrangement may be variously changed.
[0049] Referring to FIG. **3**B, a semiconductor device **10**B may include a first substrate structure S**1**, a second substrate structure S**2**, and a third substrate structure S**3**, stacked in a vertical direction. The first substrate structure S**1** and the second substrate structure S**2** may form the

memory cell array **20** of FIG. **1**. The third substrate structure S**3**, disposed between the first substrate structure S**1** and the second substrate structure S**2**, may form the peripheral circuit **30** of FIG. **1**. The first substrate structure S**1** and the second substrate structure S**2** may include first memory cell arrays MCA**1** and second memory cell arrays MCA**2**, respectively. The third substrate structure S**3** may include a row decoder DEC, a page buffer PB, and other peripheral circuits PERI, and the description of the first region R**1** described above with reference to FIG. **3**A may be applied in a similar manner.

[0050] FIG. **4** is a schematic plan view of a semiconductor device according to example embodiments. In FIG. **4**, only main components of a first memory cell region CELL**1** of the semiconductor device **100** are illustrated. FIG. **5** is a schematic cross-sectional view of a semiconductor device according to example embodiments. FIG. **5** illustrates a cross section cut along line I-I′ of FIG. **4**, including a cross section of the first memory cell region CELL**1**.

[0051] Referring to FIGS. **4** and **5**, the semiconductor device **100** may include a first substrate structure S**1** and a second substrate structure S**2**, vertically stacked. The first substrate structure S**1** may include a first memory cell region CELL**1** and a peripheral circuit region PERI. The second substrate structure S**2** may include a second memory cell region CELL**2**.

[0052] In the first substrate structure S**1**, the first memory cell region CELL**1** may be disposed on an upper end (e.g., top face or top surface) of a peripheral circuit region PERI. In example embodiments, on the other hand, the first memory cell region CELL**1** may be disposed on a lower end (e.g., bottom face or bottom surface) of the peripheral circuit region PERI. The peripheral circuit region PERI may include a base substrate **101**, circuit elements **120** disposed on the base substrate **101**, as well as circuit contact plugs **160** and circuit wiring lines **170**.

[0053] The base substrate **101** may have the upper surface extending in the X-direction and a Y-direction. The base substrate **101** may have separate element separation layers formed therein such that an active region may be defined. A portion of the active region may have source/drain regions **105** disposed therein and including an impurity. The base substrate **101** may include a semiconductor material, such as a Group IV semiconductor, a Group III-V compound semiconductor, or a Group II-VI compound semiconductor. For example, the base substrate **101** may be provided as a single crystal bulk wafer. The base substrate **101** may be described as a base semiconductor substrate, or more generally as a semiconductor substrate, which may be differentiated from other semiconductor substrates of the semiconductor device by use of ordinal numerical identifiers (e.g., first, second, or third). It will be understood that, although the terms first, second, third etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. Unless the context indicates otherwise, these terms are only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section, for example as a naming convention. Thus, a first element, component, region, layer or section discussed in one section of the specification could be termed a second element, component, region, layer or section in another section of the specification or in the claims without departing from the teachings of the present invention. In addition, in certain cases, even if a term is not described using “first,” “second,” etc., in the specification, it may still be referred to as “first” or “second” in a claim in order to distinguish different claimed elements from each other.

[0054] The circuit elements **120** may include, for example, a planar transistor. Each of the circuit elements **120** may include a circuit gate dielectric layer **122**, a spacer layer **124**, and a circuit gate electrode **125**. The source/drain regions **105** may be disposed in the base substrate **101** at both sides of the circuit gate electrode **125**.

[0055] The peripheral region insulating layer **190** may be disposed on the circuit element **120** on the base substrate **101**. The circuit contact plugs **160** may pass through the peripheral region insulating layer **190** to be connected to the source/drain regions **105**, and may include first to third circuit contact plugs **162**, **164**, and **166**, sequentially positioned from the base substrate **101**. The

circuit contact plugs **160** may allow an electrical signal to be applied to the circuit element **120**. In a region not illustrated, circuit contact plugs **160** may be connected to the circuit gate electrode **125**. The circuit wiring lines **170** may be connected to the circuit contact plugs **160**, and may include first to third circuit wiring lines **172**, **174**, and **176**, forming a plurality of layers.
[0056] The first memory cell region CELL**1**, as illustrated in FIG. **4**, may include a first substrate **201** having a cell array region CAR, which is a first region, and a cell connection region CTR, which is a second region, first gate electrodes **230** stacked on the first substrate **201**, first interlayer insulating layers **220** alternately stacked with the first gate electrodes **230**, gate separation regions SR extended while passing through a stacked structure of the first gate electrodes **230**, upper separation regions SS passing through a portion of the first gate electrodes **230**, first channels CH**1** disposed to pass through the first gate electrodes **230**, and a first cell region insulating layer **290** covering the first gate electrodes **230**. The first memory cell region CELL**1** may further include first cell contact plugs **260**, first through contact plugs **261**, first lower contact plugs **262**, first bit lines **270** and **270***a*, second lower contact plugs **264**, and first bonding pads **280**, which are wiring structures for applying a signal to the first channels CH**1** and the first gate electrodes **230**.
[0057] The cell array region CAR of the first substrate **201** may be a region in which the first gate electrodes **230** are vertically stacked and first channels CH**1** are disposed, and may be a region corresponding to the memory cell array **20** of FIG. **1**, while the cell connection region CTR may be a region in which the first gate electrodes **230** extend lengthwise by different lengths, and may correspond to a region for electrically connecting the memory cell array **20** to the peripheral circuit **30** of FIG. **1**. An item, layer, or portion of an item or layer described as extending “lengthwise” in a particular direction has a length in the particular direction and a width perpendicular to that direction, where the length is greater than the width. The cell connection region CTR may be disposed in at least one end of the cell array region CAR in at least one direction, for example, an X-direction.
[0058] The first substrate **201** may have the upper surface extending in the X-direction and a Y-direction. The first substrate **201** may include a semiconductor material, such as a Group IV semiconductor, a Group III-V compound semiconductor, or a Group II-VI compound semiconductor. For example, the Group IV semiconductor may include silicon, germanium, or silicon-germanium. For example, the first substrate **201** may be provided as a polycrystalline layer or an epitaxial layer. The first substrate **201** may be described as a memory cell region semiconductor substrate, or more generally as a semiconductor substrate, which may be differentiated from other semiconductor substrates of the semiconductor device by use of ordinal numerical identifiers (e.g., first, second, or third).
[0059] The first gate electrodes **230** may be stacked and spaced apart from each other in a direction perpendicular to the first substrate **201**, thereby forming a stacked structure together with the first interlayer insulating layers **220**. The first gate electrodes **230** may include a first lower gate electrode **231**, forming a gate of the first ground select transistor GST**1** of FIG. **2**A, first memory gate electrodes **232** to **236**, forming a plurality of first memory cells MC**1**, and first upper gate electrodes **237** and **238**, forming a gate of the first string select transistors SST**1_1** and SST**1_2**. The number of the first memory gate electrodes **232** to **236**, forming the first memory cells MC**1**, may be determined depending on capacity of the semiconductor device **100**. According to an example embodiment, the first upper and lower gate electrodes **231**, **237**, and **238** of the first string select transistors SST**1_1** and SST**1_2** and the first ground select transistor GST**1** may be provided in an amount of one or two or more, and may have the same or different structure from that of the first gate electrodes **230** of the first memory cells MC**1**. Some first gate electrodes **230**, for example, first memory gate electrodes **232** to **236**, adjacent to the first upper or lower gate electrode **231**, **237**, and **238**, may be dummy gate electrodes.
[0060] The first gate electrodes **230** may be stacked and spaced apart from each other in the cell array region CAR, and may extend lengthwise by different lengths from the cell array region CAR

into the cell connection region CTR to form a stepped staircase structure. The first gate electrodes **230** are stepped in the X-direction as illustrated in FIG. **5**, and may be disposed to be stepped in the Y-direction. Due to the stepped portion, a first gate electrode **230** in a lower portion is extended longer into the cell connection region CTR (and has a longer overall length in the X-direction) than a first gate electrode **230** in an upper portion, so the first gate electrodes **230** may provide contact regions CP exposed upwardly. The first gate electrodes **230** may be connected to the first cell contact plugs **260** in the contact regions CP, respectively.

[0061] As illustrated in FIG. **4**, the first gate electrodes **230** may be disposed to be separated from each other in the Y-direction by gate separation regions SR extended in the X-direction. The first gate electrodes **230**, between a pair of gate separation regions SR continuously extending in the X-direction among the gate separation regions SR, may form a single memory block, but a range of a memory block is not limited thereto. A portion of the first gate electrodes **230**, for example, first memory gate electrodes **232** to **236** may form a single layer in a single memory block.

[0062] The first interlayer insulating layers **220** may be disposed between the first gate electrodes **230**. The first interlayer insulating layers **220** may also be disposed to be spaced apart from each other in a direction perpendicular to the upper surface of the first substrate **201** and to extend lengthwise in the X-direction, in a manner similar to the first gate electrodes **230**. The first interlayer insulating layers **220** may contain an insulating material, such as silicon oxide or silicon nitride.

[0063] The gate separation regions SR may be disposed to pass through the first gate electrodes **230** in the cell array region CAR and the cell connection region CTR and to extend in the X-direction. The gate separation regions SR may be arranged parallel to each other. In the gate separation regions SR, a continuously extended pattern and an intermittently extended pattern may be alternately disposed in the Y-direction. However, the arrangement order, the number, and the like, of the gate separation regions SR, are not limited to those illustrated in FIG. **4**. The gate separation regions SR may pass through the entirety of the first gate electrodes **230**, stacked on the first substrate **201**, and may be connected to the first substrate **201**. The first common source line CSL**1**, described with reference to FIGS. **2**A and **2**B, may be disposed in the gate separation regions SR, and the dummy common source line may be disposed in at least a portion of the gate separation regions SR. However, the first common source line CSL**1** may be disposed in the first substrate **201** or below the first substrate **201**, according to example embodiments.

[0064] Upper separation regions SS may extend in the X-direction between the gate separation regions SR. The upper separation regions SS may be disposed in a portion of the cell connection region CTR and the cell array region CAR, to pass through a portion of first gate electrodes **230**, including the first upper gate electrodes **237** and **238**, among the first gate electrodes **230**. The first upper gate electrodes **237** and **238**, separated by the upper separation regions SS, may form different first string select lines SSL**1_1** and SSL**1_2** (See FIGS. **2**A and **2**B). The upper separation regions SS may include an insulating layer. The upper separation regions SS may separate, for example, a total of three first gate electrodes **230**, including the first upper gate electrodes **237** and **238**, from each other in the Y-direction. However, the number of the first gate electrodes **230**, separated by the upper separation regions SS, may be variously changed in example embodiments. In example embodiments, the first substrate structure S**1** may further include insulating layers separating first lower gate electrodes **231** among the first gate electrodes **230**. For example, the insulating layer may be disposed to separate first lower gate electrodes **231** in a region between the gate separation regions SR, spaced apart from each other on a straight line and arranged intermittently.

[0065] The through wiring insulating layer **295** may be disposed to pass through the first gate electrodes **230** and the first interlayer insulating layers **220** from an upper portion of the first gate electrodes **230**. The through wiring insulating layer **295** may be a region in which a wiring structure for connection of the first memory cell region CELL**1** and the peripheral circuit region

PERI is disposed. The through wiring insulating layer **295** may include an insulating material such as a silicon oxide or a silicon nitride.

[0066] The first channels CH**1** may be spaced apart from each other in rows and columns on the cell array region CAR. The first channels CH**1** may be disposed to form a grid pattern or disposed in a zigzag form in a direction. The first channel CH**1** may have a columnar shape, and may have an inclined side surface narrowing towards the first substrate **201** according to aspect ratios. In example embodiments, dummy channels may be further disposed in an end portion of the cell array region CAR, adjacent to the cell connection region CTR, and the cell connection region CTR. As for a specific structure of the first channels CH**1**, a description of the second channels CH**2** described below may be applied in a similar manner.

[0067] The first memory cell region CELL**1** may further include first cell contact plugs **260**, first through contact plugs **261**, first lower contact plugs **262**, first bit lines **270** and **270***a*, second lower contact plugs **264**, and first bonding pads **280**, which are wiring structures for electrical connection with the peripheral circuit region PERI and the second substrate structure S**2**. The wiring structures described above may include a conductive material. The wiring structures may include, for example, tungsten (W), aluminum (Al), copper (Cu), a tungsten nitride (WN), a tantalum nitride (TaN), a titanium nitride (TiN), or combinations thereof.

[0068] The first cell contact plugs **260** may pass through the first cell region insulating layer **290** to be connected to the first gate electrodes **230** in the contact regions CP. The first cell contact plugs **260** may have a cylindrical shape. In example embodiments, the first cell contact plugs **260** may have an inclined side surface narrowing towards the first substrate **201** according to aspect ratios. Thus, the first cell contact plugs **260** may have a tapered shape that tapers toward the first substrate **201**. According to example embodiments, some of the first cell contact plugs **260**, connected to certain first gate electrodes **230**, may be dummy contact plugs.

[0069] The first through contact plugs **261** may be extend vertically and may be connected to circuit wiring lines **170** of the peripheral circuit region PERI in a lower portion. The first through contact plugs **261** may pass through the through wiring insulating layer **295** and the first substrate **201** in a stacked structure of the first gate electrodes **230**, and may pass through the first cell region insulating layer **290** outside of the stacked structure of the first gate electrodes **230**. The first through contact plugs **261** may be insulated by the first substrate **201** and the side insulating layer **292**.

[0070] The first lower contact plugs **262** may be disposed on the first channels CH**1**, the first cell contact plugs **260**, and the first through contact plugs **261**.

[0071] The first bit lines **270** and **270***a* may be disposed between the first lower contact plugs **262** and the second lower contact plugs **264** at an upper end of the first lower contact plugs **262**. The first bit lines **270** and **270***a* may include first bit lines **270** connected to the first channels CH**1**, and first bit lines **270***a* connected to the first cell contact plugs **260** and the first through contact plugs **261**. The first bit lines **270**, connected to the first channels CH**1**, may correspond to the first bit lines BL**1_0** to BL**1_2** of FIG. **2**A (noting that FIG. **2**A is just a representative portion of the overall semiconductor device **100**, and does not show the same number of first bit lines as FIGS. **4** and **5**. The first bit lines **270***a*, connected to the first cell contact plugs **260** and the first through contact plugs **261**, do not correspond to the first bit lines BL**1_0** to BL**1_2** of FIG. **2**A, and may be wiring lines formed at the same level, in the same process as that of the first bit lines **270** connected to the first channels CH**1**. The first bit lines **270***a*, connected to the first cell contact plugs **260**, are illustrated as being disposed on all first gate electrodes **230**, but are not limited thereto.

[0072] The second lower contact plugs **264** are disposed on the first bit lines **270** and **270***a*, and may be connected to the first bonding pads **280** in an upper portion.

[0073] The first bonding pads **280** are disposed on the second lower contact plugs **264**, and an upper surface of the first bonding pads **280** may be exposed with respect to an upper surface of the first substrate structure S**1** through the first cell region insulating layer **290**. The first bonding pads

280 may serve as a bonding layer for bonding the first substrate structure S**1** and the second substrate structure S**2**. Bonding pads, or other pads, as described herein, are formed of conductive material and have a substantially flat, or planar, outer surface. The first bonding pads **280** may have a large planar area as compared to other wiring structures, in order to be bonded with the second substrate structure S**2** and to provide an electrical connection path thereby.

[0074] The first bonding pads **280** may be arranged in a constant pattern in each of the cell array region CAR and the cell connection region CTR. The first bonding pads **280** may be disposed at the same level in the cell array region CAR and the cell connection region CTR, and may have the same or different sizes. The first bonding pads **280** may have, for example, a circular or elliptical shape, on a plane, but are not limited thereto. The first bonding pads **280** may have a first maximum length L**1** in the cell array region CAR and may have a second maximum length L**2** in the cell connection region CTR, and the first maximum length L**1** and the second maximum length L**2** may be equal or different. Here, a 'maximum length' indicates a length corresponding to a diameter when having a circular shape on a plane, and indicates a length of the longest diagonal line when having a polygonal shape on a plane. The first maximum length L**1** may be greater than a maximum width of the first channels CH**1**. For example, the first maximum length L**1** and the second maximum length L**2** may range from several hundred nanometers to several micrometers. The first bonding pads **280** may include a conductive material, for example, copper (Cu).

[0075] In a manner similar to the first memory cell region CELL**1**, the second memory cell region CELL**2** of the second substrate structure S**2** may include a second substrate **301** having a cell array region CAR and a cell connection region CTR, second gate electrodes **330** stacked on the second substrate **301**, gate separation regions SR extending while passing through a stacked structure of the second gate electrodes **330**, upper separation regions SS passing through a portion of the second gate electrodes **330**, and second channels CH**2** disposed to pass through the second gate electrodes **330**. The second memory cell region CELL**2** may further include second interlayer insulating layers **320** alternately stacked with the second gate electrodes **330** on the second substrate **301**, a second channel region **340** in the second channels CH**2**, a second gate dielectric layer **345**, a second channel insulating layer **350**, a second channel pad **355**, and a second cell region insulating layer **390**. The second memory cell region CELL**2** may further include second cell contact plugs **360**, a second through contact plug **361**, first upper contact plugs **362**, second bit lines **370** and **370***a*, second upper contact plugs **364**, and second bonding pads **380**, which are wiring structures for applying a signal to the second channels CH**2** and the second gate electrodes **330**.

[0076] In the second substrate structure S**2**, at least gate electrodes and channels may have a symmetrical structure with those of the first substrate structure S**1**, based on an interface of the first substrate structure S**1**. With respect to each component forming the second substrate structure S**2**, the description of each component in the first memory cell region CELL**1** of the first substrate structure S**1** may be applied thereto in a similar manner, unless otherwise described. In the following discussion of the second substrate structure S**2** as a singular structure, the second substrate **301** is described as the bottom of the second substrate structure S**2**, and orientations of other components of the second substrate structure S**2** are described with that context in mind. The second substrate **301** may be described later in connection with the entire semiconductor device **100** as a top of the semiconductor device **100** or as being above certain other components.

[0077] The second substrate **301** may include a semiconductor material, such as a Group IV semiconductor, a Group III-V compound semiconductor, or a Group II-VI compound semiconductor. For example, the second substrate **301** may be provided as a single crystal bulk wafer, may be formed of a substrate the same as the base substrate **101**, and may be formed of a material having crystallinity different from that of the first substrate **201**. The second substrate **301** may be described as a memory cell region semiconductor substrate, or more generally as a semiconductor substrate, which may be differentiated from other semiconductor substrates of the semiconductor device by use of ordinal numerical identifiers (e.g., first, second, or third). The base substrate **101**, first substrate **201**, and second substrate **301** may be described as different-level substrates, such as first-level substrate, second-level substrate, and third-level substrate, where the levels refer to different vertical levels within the semiconductor device **100**.

[0078] The second gate electrodes **330**, in an amount the same as that of the first gate electrodes **230**, may be stacked on the third substrate **301**. However, the number of the second gate electrodes **330** may be variously changed in example embodiments, and the second gate electrodes **230** may be provided in an amount different from that of the first gate electrodes **230**.

[0079] The second channel region **340** may be disposed in the second channels CH**2**. In the second channel CH**2**, the second channel region **340** may have an annular form surrounding the second channel insulating layer **350**, formed therein. However, the second channel region may have a columnar shape without the second channel insulating layer **350**, such as a cylinder or a prism, according to an example embodiment. The second channel region **340** may be connected to a second epitaxial layer **307** in a lower portion of the second channel region. The second channel region **340** may contain a semiconductor material such as polycrystalline silicon or single crystalline silicon, and the semiconductor material may be a material undoped with an impurity, or a material containing p-type or n-type impurities. The second channels CH**2**, disposed on a straight line in the Y-direction between the gate separation regions SR and the upper separation region SS, may be connected to different second bit lines **370**, according to arrangement of an upper wiring structure connected to the second channel pad **355**.

[0080] The second channel pads **355** may be disposed in an upper portion of the second channel region **340** in the second channels CH**2**. The second channel pads **355** may be disposed to cover an upper surface of the second channel insulating layer **350** and to be electrically connected to the second channel region **340**. The second channel pads **355** may include, for example, doped polycrystalline silicon.

[0081] The second gate dielectric layer **345** may be disposed between the second gate electrodes **330** and the second channel region **340**. Although not specifically illustrated, the second gate dielectric layer **345** may include a tunneling layer, a charge storage layer, and a blocking layer sequentially stacked from the second channel region **340**. The tunneling layer may allow a charge to tunnel to the charge storage layer, and may include, for example, silicon oxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), silicon oxynitride (SiON), or combinations thereof. The charge storage layer may be a charge trap layer or a floating gate conductive layer. The blocking layer may include silicon oxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), silicon oxynitride (SiON), a high-k dielectric material or combinations thereof. In example embodiments, at least a portion of the second gate dielectric layer **345** may be extended in a horizontal direction along the second gate electrodes **330**.

[0082] The second epitaxial layer **307** may be disposed on the second substrate **301** in a lower end of the second channels CH**2**, and may be disposed in a side surface of at least one second gate electrode **330**. The second epitaxial layer **307** may be disposed in a recessed region of the second substrate **301**. A level of an upper surface of the second epitaxial layer **307** may be higher than a level of an upper surface of a lowermost second gate electrode **331** and may be lower than a level of a lower surface of a second gate electrode **332** located thereabove, but it is not limited to that illustrated in the drawings. In example embodiments, the second epitaxial layer **307** may be omitted. In this case, the second channel region **340** may be directly connected to the second substrate **301** or may be connected to another conductive layer on the second substrate **301**.

[0083] The second cell contact plugs **360** may only be disposed on the second gate electrodes **330** connected to the first cell contact plugs **260** of the first substrate structure S**1** at one end (e.g., in the X-direction) of the second gate electrodes **330** illustrated in the drawings. Thus, the second cell contact plugs **360**, connected to the second memory gate electrodes **332** to **336**, may be connected to the second cell contact plugs **360** at the other end of the second gate electrodes **330** in the X-direction of a stacked structure of the second gate electrodes **330**. The other end of the second gate electrodes **330** in the X-direction may also have a stepped staircase structure. In this case, the second cell contact plugs **360**, connected to the second memory gate electrodes **332** to **336**, may be connected to the peripheral circuit region PERI of the first substrate structure S**1** through first through contact plugs **261**. However, example embodiments are not limited thereto, and the second cell contact plugs **360** may be disposed on the second memory gate electrodes **332** to **336** according to example embodiments. However, in this case, the second cell contact plugs **360**, disposed on the second memory gate electrodes **332** to **336**, are not connected to the first cell contact plugs **260**, and may be connected to the peripheral circuit region PERI of the first substrate structure S**1** in a region not illustrated in the drawings.

[0084] The second through contact plugs **361** may pass through the second cell region insulating layer **390** to be connected to the second substrate **301**, and may be connected to the first through contact plugs **261** of the first memory cell region CELL**1** through the second bonding pads **380** at a lower end.

[0085] The first substrate structure S**1** and the second substrate structure S**2** may be bonded by bonding of the first bonding pads **280** and the second bonding pads **380**, for example, copper (Cu)-to-copper (Cu) bonding. The first bonding pads **280** and the second bonding pads **380** may have an area relatively larger than that of other configurations of the wiring structure, such as the first bit lines **270** and the second bit lines **370**, so reliability of the electrical connection between the first substrate structure S**1** and the second substrate structure S**2** may be improved. In example embodiments, the first substrate structure S**1** and the second substrate structure S**2**, may be bonded by hybrid bonding due to bonding of the first bonding pads **280** and the second bonding pads **380**, and dielectric-to-dielectric bonding of the first cell region insulating layer **290** and the second cell region insulating layer **390**, disposed around the first bonding pads **280** and the second bonding pads **380**.

[0086] In detail, in the semiconductor device **100**, the first bit lines **270** and the second bit lines **370**, disposed adjacent to a bonding interface of the first substrate structure S**1** and the second substrate structure S**2**, may be electrically connected to each other by a wiring structure including the first bonding pads **280** and the second bonding pads **380**. The first bit lines **270** and the second bit lines **370** may be physically and electrically connected through the second lower contact plug **264**, the first bonding pads **280** and the second bonding pads **380**, as well as the second upper contact plug **364**. In detail, the first bit line **270** and the second bit line **370**, vertically disposed in the Z-direction in parallel, may be electrically connected to each other. Moreover, for example, the first channels CH**1** and the second channels CH**2**, vertically disposed opposite each other in the Z-direction, may be electrically connected to each other, but the embodiments are not limited thereto. Thus, the first channels CH**1** and the second channels CH**2**, vertically disposed directly opposite each other, may have a structure sharing the first bit lines **270** and the second bit lines **370**. However, according to example embodiments, the first channels CH**1** may share the first bit lines **270** and the second bit lines **370** with second channels CH**2** spaced apart from each other in a horizontal direction, rather than second channels CH**2** disposed in the Z-direction directly opposite the first channels CH**1**. As described above the semiconductor device **100** may have a bit line sharing structure, and a degree of integration may be improved due to the structure described above. Though not shown in FIG. **5**, in some embodiments, each first bit line **270** in the cell array region CAR of the first substrate structure S**1** is connected to a respective second lower contact plug **264** and to a first bonding pad **280**, and each second bit line **370** in the cell array region CAR of the second substrate structure S**2** is connected to a respective second upper contact plug **364** and to a second bonding pad **380**.

[0087] In the semiconductor device **100**, at least some of the first gate electrodes **230** and the second gate electrodes **330** may be electrically connected to each other. In an example embodiment, as illustrated in a circuit diagram of FIG. **2**A, and FIG. **5**, first upper gate electrodes **237** and **238** forming a gate of the first string select transistors SST**1_1** and SST**1_2** as well as first lower gate electrodes **231** forming a gate of the first ground select transistor GST**1**, among the first gate electrodes **230**, may be electrically connected to second upper gate electrodes **337** and **338** forming a gate of the second string select transistors SST**2_1** and SST**2_2** as well as second lower gate electrodes **331** forming a gate of the second ground select transistor GST**2**, among the second gate electrodes **330**, respectively. However, according to example embodiments, only one of the string select transistors SST**1_1**, SST**1_2**, SST**2_1**, and SST**2_2** and the ground select transistors GST**1** and GST**2** may be electrically connected to each other. Alternatively, all of the first gate electrodes **230** and the second gate electrodes **330** may be individually controlled. The first gate electrodes **230** and the second gate electrodes **330**, electrically connected to each other, may be directly electrically connected at an interface through a wiring structure including the first bonding pads **280** and the second bonding pads **380**. A direct electrical connection described herein refers to a connection between two components via a continuous electrically conductive path formed therebetween. In detail, the first gate electrodes **230** and the second gate electrodes **330** may be physically and electrically connected to each other, through a first cell contact plug **260**, a first lower contact plug **262**, a first bit line **270***a*, a second lower contact plug **264**, first bonding pads **280** and second bonding pads **380**, a second upper contact plug **364**, a second bit line **370***a*, a first upper contact plug **362**, and a second cell contact plug **360**. A first gate electrode **238** and a second gate electrode **338**, in an uppermost portion, are also connected to each other through first bonding pads **280** and second bonding pads **380**, in a manner similar to the first gate electrode **237** and the second gate electrode **337** in a lower portion, in a region not illustrated.

[0088] As can be seen in FIG. **5**, in the semiconductor device **100**, a cell array region CAR of a second memory cell region CELL**2** of a second substrate structure S**2** is formed on a cell array region CAR of a first memory cell region CELL**1** of a first substrate structure S**1** in a symmetrical manner. Accordingly, in relation to the overall semiconductor device **100** having a base substrate **101** at the bottom, various plugs in a bottom substrate structure (e.g., first cell contact plugs **260** and first through contact plugs **261** of the first substrate structure S**1**) are tapered in a first direction such as downward, such that their width gets smaller in a downward direction, and various plugs in a top substrate structure (e.g., second cell contact plugs **360** and second through contact plugs **361** of the second substrate structure S**2**) are tapered in a second direction such as upward, such that their width gets larger in a downward direction. Thus, the tapering direction of certain plugs in a first (e.g., upper) substrate structure (e.g., second substrate structure S**2**) may be opposite the tapering direction of certain plugs in a second (e.g., lower) substrate structure (e.g., first substrate structure S**1**).

[0089] FIG. **6** is a layout diagram illustrating a portion of a semiconductor device according to example embodiments.

[0090] Referring to FIG. **6**, a portion of the cell array region CAR of FIG. **4** is illustrated, and arrangement on a plane, of the first channels CH**1**, the first bit lines **270**, the second lower contact plugs **264**, and the first bonding pads **280** is illustrated.

[0091] The first bit lines **270** are extended in one direction, and two first bit lines **270** may be placed over an upper portion of a single first channel CH**1**. The first bonding pads **280** may be disposed over an upper portion of the first bit lines **270**, and at least one first bonding pad **280** may be connected for each first bit line **270**. Each first bonding pad **280**, connected to a first bit line **270**, may be disposed over an upper portion of the first bit line **270** at a connection point, and may be connected to the first bit line **270** through a second lower contact plug **264**. The second lower contact plug **264** is illustrated as a quadrangle, but is not limited thereto, and may have various shapes such as an elongated, elliptical, or circular shape. Moreover, in example embodiments, the second lower contact plug **264** is extended in the Y-direction along the first bit line **270**, and may be disposed to be longer than the first bonding pad **280**.

[0092] The first bonding pads **280** may be arranged to form a diagonal pattern. The first bonding pads **280** may be disposed in parallel rows for four first channels CH**1** in the X-direction, where the parallel rows are diagonal with respect to the extension direction of the first bit lines **270**, by way of example. In the Y-direction, the first bonding pads **280** may be disposed on the adjacent first bit line **270**, shifted in the X-direction. The first bonding pads **280** have a first pitch D**1** in the X-direction, and have a second pitch D**2** in the Y-direction, a direction to which the first bit lines **270** are extended. Here, a "pitch" indicates a length between the centers of components adjacent to each other on a plane. When the components are spaced apart from each other, a "pitch" indicates a length, the sum of a maximum length of a component and a minimum distance between components. The second pitch D**2** may be greater than the first pitch D**1**, but is not limited thereto. In example embodiments, the first pitch D**1** and the second pitch D**2** may be determined in consideration of a size of the cell array region CAR, the number and a size of the first bit lines **270**, a size of the first bonding pads **280**, and the like.

[0093] FIGS. **7**A and **7**B are partially enlarged views of a semiconductor device according to example embodiments. FIGS. **7**A and **7**B illustrate an enlarged region A of FIG. **5** and an enlarged region corresponding to region A, respectively.

[0094] Referring to FIG. **7**A, arrangement of wiring structures in an upper portion of the first channel CH**1** is enlarged and illustrated. Moreover, the first channel region **240**, the first gate dielectric layer **245**, the first channel insulating layer **250**, and the first channel pad **255** of the first channel CH**1** are illustrated together. As described above with reference to FIG. **5**, the first lower contact plug **262**, the first bit line **270**, the second lower contact plug **264**, and the first bonding pad **280** are sequentially disposed on an upper portion of the first channel CH**1**.

[0095] Referring to FIG. **7**B, a structure of wiring structures according to another example embodiment is illustrated. The wiring structure may include a first lower contact plug **262**, an additional contact plug **263**, a first bit line **270**, a second lower contact plug **264**, and a first bonding pad **280**, sequentially stacked on an upper portion of the first channel CH**1**. In this example embodiment, the additional contact plug **263** may be further disposed between the first lower contact plug **262** and the first bit line **270**. Moreover, the semiconductor device according to an example embodiment may further include a bonding dielectric layer **293** surrounding the first bonding pad **280**. The bonding dielectric layer **293** may have a top surface that is coplanar with a top surface of the bonding pad **280**. The bonding dielectric layer **293** is also disposed on a lower surface of the second substrate structure S**2**, so dielectric-to-dielectric bonding may be performed thereon. The bonding dielectric layer **293** may serve as a diffusion prevention layer of the first bonding pad **280**. The bonding dielectric layer **293** may include at least one of, for example, SiO, SiN, SiCN, SiOC, SiON, and SiOCN.

[0096] FIGS. **8**A and **8**B are partially enlarged views of a semiconductor device according to example embodiments. FIGS. **8**A and **8**B illustrate an enlarged region B of FIG. **5** and an enlarged region corresponding to the region B, respectively.

[0097] Referring to FIG. **8**A, arrangement of wiring structures in an upper portion of the first cell contact plug **260** is enlarged and illustrated. As described above with reference to FIG. **5**, the first lower contact plug **262**, the first bit line **270***a*, the second lower contact plug **264**, and the first bonding pad **280** are sequentially disposed on an upper portion of the first cell contact plug **260**. The first bit line **270***a*, disposed in an upper portion of the first cell contact plug **260**, is not a layer serving as first bit lines BL**1_0** to BL**1_2** as illustrated in FIG. **2**A in a semiconductor device, but may be a layer serving as a wiring line for vertical connection.

[0098] Referring to FIG. **8**B, a structure of wiring structures according to another example embodiments is illustrated. The wiring structure may include a first lower contact plug **262**, an additional contact plug **263**, a first bit line **270***a*, a second lower contact plug **264**, and a first bonding pad **280**, sequentially stacked on an upper portion of the first cell contact plug **260**. In other words, in an example embodiment, the additional contact plug **263** may be further disposed between the first lower contact plug **262** and the first bit line **270***a*. Moreover, the semiconductor device according to an example embodiment may further include a bonding dielectric layer **293** surrounding the first bonding pad **280**.

[0099] As illustrated in FIGS. **7**A to **8**D, a structure and form of the wiring structure, disposed on an upper portion of the first channel CH**1** and the first cell contact plug **260**, may be variously changed in example embodiments. Moreover, in a single semiconductor device, structures of wiring structures disposed in an upper portion of the first channel CH**1** and an upper portion of the first cell contact plug **260** are not necessarily the same, and different wiring structures may be provided thereon.

[0100] FIGS. **9** to **12** are schematic cross-sectional views of a semiconductor device according to example embodiments.

[0101] Referring to FIG. **9**, in the semiconductor device **100***a*, as illustrated in a circuit diagram of FIG. **2**B, first memory gate electrodes **232** to **236** forming a gate of the first memory cells MC**1** as well as first lower gate electrodes **231** forming a gate of the first ground select transistor GST**1**, among the first gate electrodes **230**, may be electrically connected to second memory gate electrodes **332** to **336** forming a gate of the second memory cells MC**2** as well as second lower gate electrodes **331** forming a gate of the second ground select transistor GST**2**, among the second gate electrodes **330**, respectively. In an example embodiment, the string select transistors SST**1_1**, SST**1_2**, SST**2_1**, and SST**2_2** are separately controlled, so the first memory cells MC**1** and the second memory cells MC**2**, disposed along the first channel CH**1** and the second channel CH**2**, may be separately operated.

[0102] However, according to example embodiments, the first upper gate electrodes **237** and **238** forming a gate of the first string select transistors SST**1_1** and SST**1_2**, and the second upper gate electrodes **337** and **338** forming a gate of the second string select transistors SST**2_1** and SST**2_2** may be electrically connected to each other. In this case, the first memory cells MC**1** and the second memory cells MC**2**, disposed along the first channel CH**1** and the second channel CH**2**, may be operated in the same manner.

[0103] The first gate electrodes **230** and the second gate electrodes **330**, electrically connected to each other, may be directly connected at an interface through a wiring structure including the first bonding pads **280** and the second bonding pads **380**. Thus, all of the first and second memory gate electrodes **232** to **236** may be connected to each other through the first bonding pads **280** and the second bonding pads **380**, in regions not illustrated in the drawings.

[0104] The second cell contact plugs **360** may only be disposed on the second gate electrodes **330** connected to the first cell contact plugs **260** of the first substrate structure S**1**. Thus, the second cell contact plugs **360**, connected to the second upper gate electrodes **337** to **338**, may be connected to the second cell contact plugs **360** at the other end in the X-direction of a stacked structure of the second gate electrodes **330**, but are not limited thereto.

[0105] Referring to FIG. **10**, a region corresponding to the gate separation region SR of FIG. **4** is illustrated together with the embodiment of FIG. **5**. The first substrate structure S**1** of the semiconductor device **100***b* may further include a first source conductive layer **210** and a first source insulating layer **215**, disposed in the gate separation region SR. The second substrate structure S**2** may further include a second source conductive layer **310** and a second source insulating layer **315**, in a similar manner.

[0106] The first source conductive layer **210** may be insulated from the first gate electrodes **230** by the first source insulating layer **215**. The first source conductive layer **210** may correspond to the first common source line CSL**1** of FIGS. **2**A and **2**B. The first lower contact plug **262**, the first bit line **270**, the second lower contact plug **264**, and the first bonding pad **280** are sequentially disposed on the first source conductive layer **210**. Thus, by the first bonding pads **280** and the second bonding pads **380**, the first source conductive layer **210**, forming the first common source line CSL**1**, and the second source conductive layer **310**, forming the second common source line CSL**2**, may be electrically connected to each other.

[0107] Referring to FIG. **11**, the first substrate structure S**1** and the second substrate structure S**2** of the semiconductor device **100***c* may further include first dummy bonding pads **280**D and second dummy bonding pads **380**D, located at a level the same as a level of the first bonding pads **280** and the second bonding pads **380**, and having the same or similar shape.
[0108] The first dummy bonding pads **280**D and the second dummy bonding pads **380**D may be disposed between the first bonding pads **280** and the second bonding pads **380**, in order to enhance bonding between the first substrate structure S**1** and the second substrate structures S**2**. According to example embodiments, the first dummy bonding pads **280**D and the second dummy bonding pads **380**D may have a smaller size on a plane as compared to the first bonding pads **280** and the second bonding pads **380**, but are not limited thereto, and may have a size the same as or different from that of the first bonding pads **280** and the second bonding pads **380**. Moreover, in example embodiments, the first dummy bonding pads **280**D and the second dummy bonding pads **380**D may be arranged while forming a uniform pattern together with the first bonding pads **280** and the second bonding pads **380**.
[0109] The first dummy bonding pads **280**D and the second dummy bonding pads **380**D do not function for electrical connection. Thus, the first dummy bonding pads **280**D may not be connected to second lower contact plugs **264** in a lower portion, by way of example, and a side surface and a lower surface thereof may be disposed while being completely covered by a first cell region insulating layer **290**. Alternatively, each of the first dummy bonding pads **280**D may not be connected to any of the second lower contact plug **264**, the first lower contact plug **262**, and the first bit lines **270** and **270***a*. The second bonding pads **380**D may be disposed while being electrically insulated, in a manner similar to the description of the first dummy bonding pads **280**D.
[0110] Referring to FIG. **12**, in a semiconductor device **100***d*, first channels CH**1***a* and second channels CH**2***a* may have a U-shape. The first channel CH**1***a* may pass through a stacked structure of the first gate electrodes **230** and may have a bent form in the first substrate **201**. The first channel CH**1***a* may include a first channel region **240**, a first gate dielectric layer **245**, a first channel insulating layer **250**, and a first channel pad **255**, while the first channel region **240**, the first gate dielectric layer **245**, and the first channel insulating layer **250** may be also disposed in the U-shape. Channel separation insulating layer **296** may be further disposed between the first channel CH**1***a* and between portions of the first channel CH**1***a* having a bent form. The second channels CH**2***a* may be disposed in the U-shape, in a manner similar to the description of the first channels CH**1***a*.
[0111] Moreover, in the semiconductor device **100***d*, the first source conductive layer **210***a* and the second source conductive layer **310***a* may be disposed on one side of the first channels CH**1***a* and the second channels CH**2***a*.
[0112] FIG. **13** is a schematic cross-sectional view of a semiconductor device according to example embodiments.
[0113] Referring to FIG. **13**, the semiconductor device **200** may include a first substrate structure S**1**, a third substrate structure S**3**, and a second substrate structure S**2**, sequentially and vertically stacked. The first substrate structure S**1** may include a first memory cell region CELL**1**, the third substrate structure S**3** may include a peripheral circuit region PERI, and the second substrate structure S**2** may include a second memory cell region CELL**2**.
[0114] The first memory cell region CELL**1** may include a first substrate **201**, first gate electrodes **230**, and first channels CH**1**, and may include first cell contact plugs **260**, a first through contact plug **261**, a first lower contact plug **262**, first bit lines **270** and **270***a*, second lower contact plugs **264**, and first bonding pads **280**, which are wiring structures. The second memory cell region CELL**2**, in a manner similar to the first memory cell region CELL**1**, may include a second substrate **301**, second gate electrodes **330**, and second channels CH**2**, and may include second cell contact plugs **360**, a second through contact plug **361**, first upper contact plugs **362**, second bit lines **370** and **370***a*, second upper contact plugs **364**, and second bonding pads **380**, which are wiring

structures. The description of each component may be similarly applied to the description described above with reference to FIGS. **4** and **5**. However, in the case of the first substrate **201** and the second substrate **301**, in a manner similar to the base substrate **101**, a single crystalline layer of a semiconductor material or an epitaxial layer may be included. In the first substrate structure S**1** and the second substrate structure S**2**, the first memory cell region CELL**1** and the second memory cell region CELL**2** may have a symmetrical structure based on the third substrate structure S**3**.

[0115] The peripheral circuit region PERI may include a base substrate **101**, circuit elements **120** disposed on the base substrate **101**, circuit contact plugs **160**, including first to third circuit contact plugs **162**, **164**, and **166**, and a circuit wiring line **170**, including first and second circuit wiring lines **172** and **174**. In detail, the peripheral circuit region PERI of the semiconductor device **200** further includes circuit through contact plugs **161** passing through a base substrate **101**, as well as third bonding pads **180**A and fourth bonding pads **180**B exposed to an upper surface and a lower surface through a first peripheral region insulating layer **190** and a second peripheral region insulating layer **195**.

[0116] The circuit through contact plugs **161** may connect the third bonding pads **180**A to the fourth bonding pads **180**B, disposed on both surfaces of the base substrate **101**, respectively. The circuit through contact plugs **161** may pass through the base substrate **101** and a portion of the first peripheral region insulating layers **190**. The circuit through contact plugs **161** may be insulated from the base substrate **101** by a substrate insulating layer **140** disposed on a portion of a side surface. The circuit through contact plugs **161** may have a shape in which a width of a lower portion is greater than a width of an upper portion, but they are not limited thereto.

[0117] The third bonding pads **180**A and the fourth bonding pads **180**B are disposed on both surfaces of the third substrate structure S**3**, respectively, and may be connected to each other through the circuit through contact plugs **161**, the second circuit wiring lines **174**, and the third circuit contact plugs **166**. However, a structure of circuit wiring structures, disposed between the third bonding pads **180**A and the fourth bonding pads **180**B may be variously changed in example embodiments. The fourth bonding pads **180**B may be disposed to be in contact with an upper surface of the base substrate **101**. The third bonding pads **180**A and the fourth bonding pads **180**B may include, for example, copper (Cu).

[0118] The third bonding pads **180**A may be bonded to the first bonding pads **280** of the first substrate structure S**1**, and the fourth bonding pads **180**B may be bonded to the second bonding pads **380** of the second substrate structure S**2**. Thus, the third bonding pads **180**A are electrically connected to the first bit lines **270** and the first cell contact plugs **260**, and the fourth bonding pads **180**B may be electrically connected to the second bit lines **370** and the second cell contact plugs **360**. Thus, the first substrate structure S**1**, the second substrate structure S**2**, and the third substrate structure S**3** may be electrically connected to each other through the third bonding pads **180**A and the fourth bonding pads **180**B.

[0119] The first bit lines **270** and the second bit lines **370** of the first substrate structure S**1** and the second substrate structure S**2** may be physically and electrically connected through a second lower contact plug **264**, first bonding pads **280** and third bonding pads **180**A, third circuit contact plugs **166**, second circuit wiring lines **174**, circuit through contact plugs **161**, fourth bonding pads **180**B and second bonding pads **380**, and a second upper contact plug **364**. At least a portion of the first gate electrodes **230** and the second gate electrodes **330** of the first substrate structure S**1** and the second substrate structure S**2** may be electrically connected to each other. In detail, the first gate electrodes **230** and the second gate electrodes **330** may be physically and electrically connected to each other, through a first cell contact plug **260**, a first lower contact plug **262**, a first bit line **270***a*, a second lower contact plug **264**, first bonding pads **280** and third bonding pads **180**A, third circuit contact plugs **166**, second circuit wiring lines **174**, circuit through contact plugs **161**, fourth bonding pads **180**B and second bonding pads **380**, a second upper contact plug **364**, a second bit line **370***a*, a first upper contact plug **362**, and a second cell contact plug **360**.

[0120] FIGS. **14**A to **14**H are schematic cross-sectional views illustrating a method for manufacturing a semiconductor device according to example embodiments. FIGS. **14**A to **14**H illustrate a region corresponding to FIG. **5**.
[0121] Referring to FIG. **14**A, circuit elements **120** and circuit wiring structures are formed on the base substrate **101**, thereby forming a peripheral circuit region PERI.
[0122] First, a circuit gate dielectric layer **122** and a circuit gate electrode **125** may be sequentially formed on the base substrate **101**. The circuit gate dielectric layer **122** and the circuit gate electrode **125** may be formed using atomic layer deposition (ALD) or chemical vapor deposition (CVD). The circuit gate dielectric layer **122** may be formed of silicon oxide, and the circuit gate electrode layer **125** may be formed of at least one of polycrystalline silicon or metal silicide, but an example embodiment is not limited thereto. Then, the spacer layer **124** and the source/drain regions **105** may be formed on both side walls of the circuit gate dielectric layer **122** and the circuit gate electrode **125**. According to example embodiments, the spacer layer **124** may be formed of a plurality of layers. Then, the source/drain regions **105** may be formed by performing an ion implantation process.
[0123] The circuit contact plugs **160** of the circuit wiring structures may be provided by forming a portion of the peripheral region insulating layer **190**, etching and removing a portion and embedding a conductive material. The circuit wiring lines **170** may be provided by depositing and patterning a conductive material, by way of example.
[0124] The peripheral region insulating layer **190** may be formed of a plurality of insulating layers. The peripheral region insulating layer **190** may be ultimately provided to cover the circuit elements **120** and the circuit wiring structures, by forming a portion in respective operations for formation of the circuit wiring structures and forming a portion in an upper portion of the third circuit wiring line **176**.
[0125] Referring to FIG. **14**B, for formation of the first memory cell region CELL**1**, a first substrate **201** may be formed in an upper portion of the peripheral region insulating layer **190**. Then, sacrificial layers **225** and first interlayer insulating layers **220** are alternately stacked on the first substrate **201**, and a portion of the sacrificial layers **225** and the first interlayer insulating layers **220** may be removed to allow the sacrificial layers **225** to be extended by different lengths.
[0126] The first substrate **201** may be formed on the peripheral region insulating layer **190**. The first substrate **201** may be formed of polycrystalline silicon, for example, and may be formed using a CVD process. The polycrystalline silicon, forming the first substrate **201**, may contain an impurity. The first substrate **201** may be formed smaller than the base substrate **101**, but is not limited thereto.
[0127] The sacrificial layers **225** may be a layer to be replaced with first gate electrodes **230** through a subsequent process. The sacrificial layers **225** may be formed of a material to be etched with etching selectivity for the first interlayer insulating layers **220**. For example, the first interlayer insulating layer **220** may include at least one of silicon oxide and silicon nitride, and the sacrificial layers **225** may include a material selected from silicon, silicon oxide, silicon carbide, and silicon nitride, different from that of the first interlayer insulating layer **220**. In example embodiments, all of thicknesses of the first interlayer insulating layers **220** may not be the same.
[0128] Then, in order to allow sacrificial layers **225** in an upper portion to be extended less than sacrificial layers **225** in a lower portion, a photolithography process and an etching process for the sacrificial layers **225** and the first interlayer insulating layers **220** may be repeatedly performed. Thus, the sacrificial layers **225** may have a stepped form. In example embodiments, the sacrificial layers **225** may be formed to have a relatively thick thickness at an end portion (e.g., in the Z-direction), and a process therefor may be further performed. Then, a first cell region insulating layer **290** covering an upper portion of a stacked structure of the sacrificial layers **225** and the first interlayer insulating layers **220** may be provided.
[0129] Referring to FIG. **14**C, a through wiring insulating layer **295** and first channels CH**1**,

passing through a stacked structure of the sacrificial layers **225** and the first interlayer insulating layers **220**, are provided.

[0130] The through wiring insulating layer **295** may be formed by forming an opening by removing a portion of the sacrificial layers **225** and the first interlayer insulating layers **220** using a mask pattern, and depositing an insulating material filling the opening. According to example embodiments, during formation of the opening, a portion of the first substrate **201** may be recessed. For formation of the first channels CH**1**, first, the stacked structure may be an isotropically etched to form channel holes. Due to a height of the stacked structure, a side wall of the channel holes may not be perpendicular to an upper surface of the first substrate **201**, and so the channel holes and first channels CH**1** may have a tapered shape that tapers toward the first substrate **201**. In example embodiments, the channel holes may be formed to recess a portion of the first substrate **201**.

[0131] Then, the first epitaxial layer **207**, the first channel region **240**, the first gate dielectric layer **245**, the first channel insulating layer **250**, and the first channel pads **255** are formed in the channel holes, thereby forming the first channels CH**1**. The first epitaxial layer **207** may be formed using a selective epitaxial growth (SEG) process. The first epitaxial layers **207** may include a single layer or a plurality of layers. The first epitaxial layers **207** may contain polycrystalline silicon (Si), monocrystalline Si, polycrystalline germanium (Ge) or monocrystalline Ge that are doped with or do not include an impurity. The first gate dielectric layer **245** may be formed to have a uniform thickness using ALD or CVD. In the operation described above, at least a portion, vertically extended along the first channel region **240**, of the first gate dielectric layer **245**, may be provided. The first channel region **240** may be formed on the first gate dielectric layer **245** in the first channels CH**1**. The first channel insulating layer **250** may be formed to fill the first channels CH**1**, and may be an insulating material. However, according to example embodiments, rather than the first channel insulating layer **250**, a conductive material may fill a space of the first channel region **240**. The first channel pads **255** may be formed of a conductive material, for example, polycrystalline silicon.

[0132] Referring to FIG. **14**D, openings, passing through a stacked structure of the sacrificial layers **225** and the first interlayer insulating layers **220**, are provided, and the sacrificial layers **225** may be removed through the openings.

[0133] The openings may be provided in the form of a trench, extended in the X-direction in a region, not illustrated. The sacrificial layers **225** may be removed selectively with respect to the first interlayer insulating layers **220**, using, for example, wet etching. Thus, a portion of side walls of the first channels CH**1** and the through wiring insulating layer **295** may be exposed between the first interlayer insulating layers **220**.

[0134] Referring to FIG. **14**E, first gate electrodes **230** may be provided in a region from which the sacrificial layers **225** are removed.

[0135] A conductive material is embedded in the region, from which the sacrificial layers **225** are removed, to provide the first gate electrodes **230**. The first gate electrodes **230** may contain metal, polycrystalline silicon or metal silicide material. In example embodiments, before the first gate electrodes **230** are provided, when a region, horizontally extending on the first substrate **201** along the first gate electrodes **230**, of the first gate dielectric layer **245**, is provided, the region described above may be provided first.

[0136] Then, in a region not illustrated, in a manner similar to an example embodiment of FIG. **10**, first source insulating layers **215** and first source conductive layers **210** may be provided in the openings. The first source insulating layers **215** may be provided in the form of a spacer, by forming an insulating material and removing the insulating material from the first substrate **201** to allow an upper surface of the first substrate **201** to be exposed. The first source conductive layer **210** may be formed by depositing a conductive material between the first source insulating layers **215**. The first gate electrodes **230** may be spaced apart from each other by a predetermined distance in the Y-direction by the first source insulating layers **215** and the first source conductive layer **210**.

However, formation of the first source conductive layers **210** is not necessary, and may be omitted according to example embodiments. In this case, a layer serving a function of a source conductive layer may be formed in the first substrate **201**.

[0137] Referring to FIG. **14**F, the first cell contact plugs **260**, the first through contact plugs **261**, the first lower contact plugs **262**, the first bit lines **270** and **270***a*, the second lower contact plugs **264**, and the first bonding pads **280**, which are wiring structures, may be provided on the first gate electrodes **230**.

[0138] The first cell contact plugs **260** may be formed by etching the first cell region insulating layer **290** to form a contact hole on contact regions CP, and embedding a conductive material. Then, an insulating layer, forming the first cell region insulating layer **290** while covering an upper surface of the first cell contact plugs **260**, may be provided. Each of the first cell contact plugs **260** may be continuously formed pillars extending between first lower contact plugs **262** and a respective gate electrode **230**.

[0139] The first through contact plugs **261** may be provided by forming a through hole passing through the through wiring insulating layer **295** and the first substrate **201**, forming a side insulating layer **292** on an exposed side wall of the first substrate **201**, and then depositing a conductive material. The first lower contact plugs **262** may be formed by etching the first cell region insulating layer **290** and depositing a conductive material on the first channel pads **155**, the first cell contact plugs **260**, and the first through contact plugs **261**. Each of the first through contact plugs **261** may be continuously formed pillars extending between first lower contact plugs **262** and peripheral region insulating layer **190**, and may have a length longer than the stack of first gate electrodes **230** in the Z-direction.

[0140] The first bit lines **270** and **270***a* may be formed through deposition and patterning processes of a conductive material, or by forming a single layer, an insulating layer forming the first cell region insulating layer **290**, and then patterning it and depositing a conductive material. The second lower contact plugs **264** may be formed by etching the first cell region insulating layer **290** and depositing a conductive material on the first bit lines **270** and **270***a*.

[0141] The first bonding pads **280** may be formed through, for example, deposition and patterning processes of a conductive material on the second lower contact plugs **264**. An upper surface of the first bonding pads **280** may be exposed through the first cell region insulating layer **290**, and the first bonding pads may form a portion of an upper surface of the first substrate structure S**1**. According to example embodiments, the upper surface of the first bonding pads **280** may be provided in the form further protruding upwardly, as compared to an upper surface of the first cell region insulating layer **290**. However, in some embodiments, the upper surface of the first bonding pads **280** is provided to be coplanar with an upper surface of the first cell region insulating layer **290**. Due to the operation described above, a first memory cell region CELL**1** is completed, and the first substrate structure S**1** may be ultimately prepared.

[0142] Referring to FIG. **14**G, the second substrate structure S**2** may be provided.

[0143] The second substrate structure S**2** may be manufactured using operations the same as those of the first memory cell region CELL**1** of the first substrate structure S**1**, described above with reference to FIGS. **14**B to **14**F.

[0144] According to example embodiments, the first bit lines **270** and the second bit lines **370** are provided after formation of the first gate electrodes **230** and the second gate electrodes **330**. Thus, even when the first bit lines **270** and the second bit lines **370** are formed of a material, relatively vulnerable to a high-temperature process, for example, copper (Cu), the bit lines may be formed without restriction of a process.

[0145] Referring to FIG. **14**H, the second substrate structure S**2** may be bonded to the first substrate structure S**1**.

[0146] The first substrate structure S**1** and the second substrate structure S**2** may be connected to each other by bonding the first bonding pads **280** and the second bonding pads **380**, for example by applying a pressure. The second substrate structure S**2** may be bonded to the first substrate structure S**1** by inverting the second substrate structure to allow the second bonding pads **380** to face downwardly. The first substrate structure S**1** and the second substrate structure S**2** may be directly bonded without intervention of an adhesive such as a separate adhesive layer. For example, bonding of the first bonding pads **280** and the second bonding pads **380** at an atomic level may be provided by applying a pressure as described above. In this manner, the first bonding pads **280** and the second bonding pads **380** contact each other. According to example embodiments, before bonding, in order to enhance a bonding force, a surface treatment process such as a hydrogen plasma treatment may be further performed on an upper surface of the first substrate structure S**1** and a lower surface of the second substrate structure S**2**.

[0147] In example embodiments, as illustrated in FIGS. **7**B and **8**B, when the bonding dielectric layer **293** is disposed in an upper portion of the first cell region insulating layer **290** and the second substrate structure S**2** also has the same layer, a bonding force may be further secured due to not only bonding between the first bonding pads **280** and the second bonding pads **380**, but also dielectric bonding between the bonding dielectric layer **293** and a bonding dielectric layer disposed thereon. In this manner, the bonding dielectric layer **293** of the first substrate structure S**1** contacts the bonding dielectric layer **293** of the second substrate structure S**2**. Due to the bonding process described above, the semiconductor device **100** of FIG. **5** may be ultimately manufactured.

[0148] FIGS. **15**A to **15**G are schematic cross-sectional views illustrating a method for manufacturing a semiconductor device according to example embodiments. FIGS. **15**A to **15**G illustrate a region corresponding to FIG. **13**.

[0149] Referring to FIG. **15**A, a first substrate structure S**1** may be provided.

[0150] The first substrate structure S**1** may be manufactured by forming the first memory cell region CELL**1**, by performing operations the same as those described above with reference to FIGS. **14**B to **14**F.

[0151] Referring to FIG. **15**B, a second substrate structure S**2** may be provided.

[0152] The second substrate structure S**2** may also be manufactured by forming the second memory cell region CELL**2**, by performing operations the same as those described above with reference to FIGS. **14**B to **14**F.

[0153] Referring to FIG. **15**C, for formation of the third substrate structure S**3** including the peripheral circuit region PERI, circuit elements **120** and circuit through contact plugs **161** may be provided on the base substrate **101**.

[0154] The circuit elements **120** may be formed using operations the same as described with reference to FIG. **14**A. After the circuit elements **120** are provided, a first peripheral region insulating layer **190** may be formed on the circuit elements **120**.

[0155] Forming the circuit through contact plugs **161** may include forming contact holes by removing a portion of the first peripheral region insulating layer **190** and the base substrate **101** from an upper surface of the first peripheral region insulating layer **190**. Then, a substrate insulating layer **140** may be formed on a side wall and a lower surface of the base substrate **101**, having been exposed through the contact holes. Then, a conductive material is embedded in the contact holes to provide the circuit through contact plugs **161**.

[0156] Referring to FIG. **15**D, circuit wiring structures forming the peripheral circuit region PERI may be provided.

[0157] The circuit contact plugs **160** of the circuit wiring structures may be provided by forming a portion of the first peripheral region insulating layer **190**, etching and removing a portion and embedding a conductive material. The circuit wiring lines **170** may be provided by depositing and patterning a conductive material, by way of example. Then, third bonding pads **180**A may be formed on the second circuit contact plugs **164**.

[0158] Referring to FIG. **15**E, the third substrate structure S**3** may be bonded to the first substrate structure S**1**.

[0159] The first substrate structure S**1** and the third substrate structure S**3** may be connected to each other by bonding the first bonding pads **280** and the third bonding pads **180**A by applying a pressure. The third substrate structure S**3** in an inverted form may be bonded to the first substrate structure S**1** to allow the base substrate **101** to face upwardly. The first substrate structure S**1** and the third substrate structure S**3** may be directly bonded without intervention of an adhesive such as a separate adhesive layer. In example embodiments, as illustrated in FIGS. **7**B and **8**B, when the bonding dielectric layer **293** is disposed in an upper portion of the first cell region insulating layer **290** and the third substrate structure S**3** also has the same dielectric layer, a bonding force may be further secured due to dielectric bonding.
[0160] Referring to FIG. **15**F, the base substrate **101** is removed by a predetermined thickness from an unbonded surface of the base substrate **101**, that is, an upper surface of a bonding structure of the first substrate structure S**1** and the third substrate structure S**3**, to form the third bonding pads **180**B.
[0161] A portion of the base substrate **101** may be removed to expose the circuit through contact plugs **161** using a grinding process, a planarization process, or the like. During the removal process, the substrate insulating layer **140**, a bottom surface of the circuit through contact plugs **161**, having been formed on an upper surface in the FIG. **15**F, may be removed. Then, third bonding pads **180**B may be formed on the circuit through contact plugs **161**. A second peripheral region insulating layer **195**, surrounding the third bonding pads **180**B, may be provided. However, according to example embodiments, the second peripheral region insulating layer **195** may be provided first before the third bonding pads **180**B are provided.
[0162] Referring to FIG. **15**G, the second substrate structure S**2** may be bonded to a bonding structure the first substrate structure S**1** and the third substrate structure S**3**.
[0163] The bonding structure the first substrate structure S**1** and the third substrate structure S**3**, and the second substrate structure S**2** may be connected to each other by bonding the fourth bonding pads **180**B and the second bonding pads **380** by applying a pressure. The second substrate structure S**2** in an inverted form may be bonded to the bonding structure the first substrate structure S**1** and the third substrate structure S**3**, to allow the second substrate **301** to face upwardly. The third substrate structure S**3**, of the bonding structure the first substrate structure S**1** and the third substrate structure S**3**, and the second substrate structure S**2** may be directly bonded without intervention of an adhesive such as a separate adhesive layer. In example embodiments, the second structure S**2** and the third structure S**3** may have a bonding dielectric layer on a bonding surface. In this case, a bonding force may be further secured by dielectric bonding.
[0164] Due to the bonding process described above, the semiconductor device **200** of FIG. **13** may be ultimately manufactured.
[0165] Though not shown in FIG. **5** or **13**, one or more of the external-facing surfaces of the semiconductor substrates facing away from the semiconductor device **100** or **200** may include external connection terminals thereon (e.g., solder bumps or balls) that connect to circuitry within the semiconductor substrate to allow electrical communication between the internal circuitry of the semiconductor device **100** or **200** and a device outside of the semiconductor device. For example, the semiconductor device **100** or **200** may be described as a semiconductor chip, which may be a part of a semiconductor package, for example, that includes a package substrate connected to the semiconductor chip by using the external connection terminals.
[0166] FIG. **16** is a block diagram illustrating an electronic device including a semiconductor device according to example embodiments.
[0167] Referring to FIG. **16**, an electronic device **1000** according to an example embodiment may include a communications unit **1010**, an input unit **1020**, an output unit **1030**, a memory **1040**, and a processor **1050**.
[0168] The communication unit **1010** may include a wired/wireless communications module such as a wireless Internet module, a local communications module, a global positioning system (GPS)

module, or a mobile communications module. The wired/wireless communications module included in the communications unit **1010** may be connected to an external communications network based on various communications standards to transmit and receive data. The input unit **1020** may include a mechanical switch, a touchscreen, a voice recognition module, and the like, as a module provided for a user to control operations of the electronic device **1000**, and may further include various sensor modules to which a user may input data. The output unit **1030** may output information processed by the electronic device **1000** in an audio or video format, and the memory **1040** may store a program for processing or control of the processor **1050**, or data. The memory **1040** may include one or more semiconductor devices according to various example embodiments as described above with reference to FIGS. **4** to **13**, and may be embedded in the electronic device **1000** or may communicate with the processor **1050** through a separate interface. The processor **1050** may control operations of each component included in the electronic device **1000**. The processor **1050** may perform control and processing associated with a voice call, a video call, data communications, and the like, or may conduct control and processing for multimedia reproduction and management. Moreover, the processor **1050** may process the input from a user via the input unit **1020** and output the result thereof through the output unit **1030**, and may store data, required for controlling an operation of the electronic device **1000**, in the memory **1040** or retrieve the data from the memory **1040**.

[0169] As set forth above, according to example embodiments of the present inventive concept, due to a bit line sharing structure in which two or more substrate structures are bonded, a semiconductor device with an improved degree of integration may be provided.

[0170] Moreover, a method for manufacturing a semiconductor device with improved reliability by forming a bit line sharing structure by bonding two or more substrate structures using a bonding pad.

[0171] While example embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present disclosure, as defined by the appended claims.

Claims

1. A method for manufacturing a semiconductor device, comprising: forming a first substrate structure; forming a second substrate structure; and bonding the second substrate structure to the first substrate structure, wherein the forming of the first substrate structure includes: forming a first stacked structure of first sacrificial layers and first interlayer insulating layers alternately stacked on a first substrate; forming first channels passing through the first stacked structure; forming first openings passing through the first stacked structure; removing the first sacrificial layers through the first openings; forming first gate electrodes by filling regions from which the first sacrificial layers are removed with a conductive material; forming first bit lines connected to the first channels; and forming first bonding pads on the first bit lines to be electrically connected to the first bit lines so that each respective first bit line is electrically connected to a respective first bonding pad that vertically overlaps the respective first bit line and one or more adjacent first bit lines, wherein the forming of the second substrate structure includes: forming a second stacked structure of second sacrificial layers and second interlayer insulating layers alternately stacked on a second substrate; forming second channels passing through the second stacked structure; forming second openings passing through the second stacked structure; removing the second sacrificial layers through the second openings; forming second gate electrodes by filling regions from which the second sacrificial layers are removed with a conductive material; forming second bit lines connected to the second channels; and forming second bonding pads on the second bit lines to be electrically connected to the second bit lines so that each respective second bit line is electrically connected to a respective second bonding pad that vertically overlaps the respective second bit line and one or more adjacent second bit lines, wherein the bonding of the second substrate structure to the first substrate structure includes bonding the first bonding pads and the second bonding pads, wherein the first bit lines are electrically connected to the second bit lines, respectively, through the first bonding pads and the second bonding pads, and wherein: the first bonding pads are arranged in rows that are diagonal with respect to an extension direction of the first bit lines in a plan view; and the second bonding pads are arranged in rows that are diagonal with respect to an extension direction of the second bit lines in the plan view.

2. The method for manufacturing a semiconductor device of claim 1, wherein the bonding of the first bonding pads and the second bonding pads is performed by applying a pressure without intervention of an adhesive.

3. The method for manufacturing a semiconductor device of claim 1, wherein each of the first bit lines is electrically connected to each of the second bit lines arranged to vertically overlap each other.

4. The method for manufacturing a semiconductor device of claim 1, wherein: the forming of the first substrate structure further includes forming lower contact plugs between the first channels and the first bit lines, and the forming of the second substrate structure further includes forming upper contact plugs between the second channels and the second bit lines.

5. The method for manufacturing a semiconductor device of claim 1, wherein: the forming of the first substrate structure further includes forming lower contact plugs between the first bonding pads and the first bit lines, and the forming of the second substrate structure further includes forming upper contact plugs between the second bonding pads and the second bit lines.

6. The method for manufacturing a semiconductor device of claim 1, wherein: the forming of the first substrate structure further includes removing a portion of the first sacrificial layers to allow the first gate electrodes to be extended by different lengths in the forming of the first gate electrodes, to provide contact regions at ends of the first gate electrodes, and the forming of the second substrate structure further includes removing a portion of the second sacrificial layers to allow the second gate electrodes to be extended by different lengths in the forming of the second gate electrodes, to provide contact regions at ends of the second gate electrodes.

7. The method for manufacturing a semiconductor device of claim 6, wherein the forming of the first substrate structure further includes: forming first cell contact plugs extending perpendicular to a surface of the first substrate and connected to the first gate electrodes, respectively, in the contact regions of the first gate electrodes, and forming third bonding pads disposed on the first cell contact plugs so that the first cell contact plugs are between the first gate electrodes and the third bonding pads, respectively, wherein the forming of the second substrate structure further includes: forming second cell contact plugs extending perpendicular to a surface of the second substrate and connected to the second gate electrodes, respectively, in the contact regions of the second gate electrodes, and forming fourth bonding pads disposed on the second cell contact plugs so that the second cell contact plugs are between the second gate electrodes and the fourth bonding pads, respectively, and wherein the third bonding pad and the fourth bonding pad are bonded to each other.

8. The method for manufacturing a semiconductor device of claim 7, wherein the third bonding pads and the fourth bonding pads electrically connect the first cell contact plugs to the second cell contact plugs.

9. The method for manufacturing a semiconductor device of claim 1, wherein the first gate electrodes and the second gate electrodes as well as the first channels and the second channels are disposed symmetrically based on an interface between the first substrate structure and the second substrate structure.

10. The method for manufacturing a semiconductor device of claim 1, wherein the forming of the first substrate structure further includes: forming a base substrate; forming circuit elements on the base substrate; and forming the first substrate on the circuit elements.

11. The method for manufacturing a semiconductor device of claim 10, wherein the base substrate and the second substrate include a single crystalline layer, and the first substrate includes a polycrystalline layer or an epitaxial layer.

12. The method for manufacturing a semiconductor device of claim 1, wherein: the forming of the first substrate structure further includes forming a first dielectric layer surrounding the first bonding pads, and the forming of the second substrate structure further includes forming a second dielectric layer surrounding the second bonding pads, wherein the bonding of the second substrate structure to the first substrate structure includes bonding the first dielectric layer and the second dielectric layer.

13. A method for manufacturing a semiconductor device, comprising: forming a first substrate structure; forming a second substrate structure; and bonding the second substrate structure to the first substrate structure, wherein the forming of the first substrate structure includes: forming a first stacked structure of first sacrificial layers and first interlayer insulating layers alternately stacked on a first substrate; forming first channels passing through the first stacked structure; forming first openings passing through the first stacked structure; removing the first sacrificial layers through the first openings; forming first gate electrodes by filling regions from which the first sacrificial layers are removed with a conductive material; forming first bit lines connected to the first channels; and forming first bonding pads on the first bit lines to be electrically connected to the first bit lines, respectively, wherein the forming of the second substrate structure includes: forming a second stacked structure of second sacrificial layers and second interlayer insulating layers alternately stacked on a second substrate; forming second channels passing through the second stacked structure; forming second openings passing through the second stacked structure; removing the second sacrificial layers through the second openings; forming second gate electrodes by filling regions from which the second sacrificial layers are removed with a conductive material; forming second bit lines connected to the second channels; and forming second bonding pads on the second bit lines to be electrically connected to the second bit lines, respectively, wherein the bonding of the second substrate structure to the first substrate structure includes bonding the first bonding pads and the second bonding pads, wherein the first bit lines are electrically connected to the second bit lines, respectively, through the first bonding pads and the second bonding pads, and wherein each respective first bit line, first bonding pad, second bit line, and second bonding pad are electrically connected and vertically aligned.

14. The method for manufacturing a semiconductor device of claim 13, wherein: each of the first bonding pads vertically overlaps the respective first bit line and one or more adjacent first bit lines; and each of the second bonding pads vertically overlaps the respective second bit line and one or more adjacent second bit lines.

15. The method for manufacturing a semiconductor device of claim 13, wherein: the first bonding pads are arranged in rows that are diagonal with respect to an extension direction of the first bit lines in a plan view; and the second bonding pads are arranged in rows that are diagonal with respect to an extension direction of the second bit lines in the plan view.

16. The method for manufacturing a semiconductor device of claim 13, wherein, in a direction perpendicular to an upper surface of the first substrate, the first bit lines are disposed between the first channels and the first bonding pads, and the first channels extend between the first substrate and the first bit lines.

17. The method for manufacturing a semiconductor device of claim 16, wherein, in a direction perpendicular to an upper surface of the second substrate, the second bit lines are disposed between the second channels and the second bonding pads, and the second channels extend between the second substrate and the second bit lines.

18. A method for manufacturing a semiconductor device, comprising: forming a first substrate structure including a first substrate, first memory cells disposed on the first substrate, first bit lines disposed on the first memory cells and connected to the first memory cells, and first bonding pads disposed on the first bit lines to be connected to the first bit lines, respectively; forming a second substrate structure including a second substrate, second memory cells disposed on the second substrate, second bit lines disposed on the second memory cells and connected to the second memory cells, and second bonding pads disposed on the second bit lines to be connected to the second bit lines, respectively; and bonding the second substrate structure to the first substrate structure by bonding the first bonding pads and the second bonding pads, wherein the first bit lines are electrically connected to the second bit lines, respectively, through the first bonding pads and the second bonding pads, and wherein the first bonding pads and second bonding pads are vertically between the first bit lines and the second bit lines.

19. The method for manufacturing a semiconductor device of claim 18, wherein each respective first bit line, first bonding pad, second bit line, and second bonding pad, that are electrically connected, vertically overlap.

20. The method for manufacturing a semiconductor device of claim 18, wherein: the first bonding pads are arranged in rows that are diagonal with respect to an extension direction of the first bit lines in a plan view; and the second bonding pads are arranged in rows that are diagonal with respect to an extension direction of the second bit lines in the plan view.